MURRAY ACOUSTICS

The Grange, Bank Lane Roehampton, London

Noise Impact Assessment

21157-RP-NIA-01

30 September 2026

Knight Frank LLP
55 Baker Street
London
W1U 8AN

Document Ref.	Rev.	Date	Author	Comments
21157-RP-NIA-01	-	30 September 2026	**Jamie Murray MIOA** Director	-

Disclaimer

This report has been prepared by Murray Acoustics Ltd for the sole benefit of the named client and may not be relied upon by any third party without the prior written consent of Murray Acoustics Ltd.

The conclusions and recommendations contained in this report are based on the information available at the time of writing, including any data, drawings, or specifications provided by the client or other parties. Murray Acoustics Ltd accepts no responsibility for any inaccuracies in information provided by others.

The assessment is based on the specific conditions, equipment selections, and site layouts described herein. Any changes to these may require the assessment to be reviewed and updated.

This report does not purport to provide legal advice. The findings represent the professional opinion of Murray Acoustics Ltd and do not constitute a guarantee of compliance with any statutory or regulatory requirements.

Murray Acoustics Ltd 2026. All rights reserved.

Table of Contents

1.0 Introduction 3
1.1 Overview 3
1.2 Site Description 3
1.3 Development Proposals 5
1.4 Assessment Scope 6
2.0 Assessment Guidance and Requirements 8
2.1 Permitted Development Requirements 8
2.2 BS 8233:2014 8
2.3 BS 4142:2014+A1:2019 9
3.0 Environmental Noise Survey 10
3.1 Procedure 10
3.2 Noise Climate 10
3.3 Measurement Positions 11
3.4 Equipment 11
3.5 Results 12
4.0 Commercial Noise Assessment 13
4.1 Commercial Premises Surrounding the Site 13
4.2 Assessment Approach 14
4.3 Criteria 14
4.4 Internal Noise Levels with Windows Open 14
4.5 Non-Glazed Elements 15
4.6 Glazed Elements 15
4.7 Ventilation 17
4.7.1 Normal Ventilation 17
4.7.2 Overheating (Approved Document O) 18
4.8 Existing Building Services Plant 19
4.9 External Amenity 19
5.0 Conclusion 20
Appendix A – Environmental Noise Survey Results 22

1.0 Introduction

1.1 Overview

Murray Acoustics Ltd has been commissioned by Knight Frank LLP, on behalf of the Bank of England, to undertake a noise impact assessment of the proposed change of use of The Grange, Bank Lane, Roehampton, London, SW15 5XZ from offices (Use Class E) to 48No residential apartments (Use Class C3).

The change of use is being progressed under permitted development rights, and the assessment has been prepared to support the prior approval application to the London Borough of Wandsworth under Class MA of Part 3 of Schedule 2 to the Town and Country Planning (General Permitted Development) (England) Order 2015, as amended, in respect of the impacts of noise from commercial premises on the intended occupiers of the development.

This report sets out the assessment methodology, the results of a seven-day environmental noise survey at the building, the findings of the assessment and the recommended mitigation measures.

1.2 Site Description

The Grange is a five-storey building constructed in 1908 as the Bank of England's records office and most recently in use as multi-let offices. It is of solid masonry construction with an H-shaped plan, a central entrance and a clock tower, and stands at the south-western corner of the former Bank of England Sports Ground, a site of approximately 28 acres in Roehampton, south-west London, some 500 m from the entrance to Richmond Park. The site location is shown in Figure 1.1 and an aerial view of the wider sports ground in Figure 1.2.

The building fronts Bank Lane to the south-west, beyond which lie the detached dwellings of Bank Lane and the residential streets of Roehampton and Barnes. To the north of the building lies the sports pavilion (clubhouse) of the sports ground, a building which historically contained three function rooms and a café, together with an adjoining sports hall and swimming pool, with three squash courts to the east of the building; the pavilion is understood to be no longer in use as a function or leisure venue and the squash courts are disused.

Beyond the pavilion the sports ground comprises extensive grass playing fields, cricket squares and grass and hard tennis courts, extending some 500 m to the north. To the south-east of the building, beyond its car park, is an all-weather (2G) sports pitch. To the east of the sports ground lies the National Tennis Centre operated by the Lawn Tennis Association, and beyond that the Roehampton Club golf course.

The sports ground was sold by the Bank of England in 2020 and is now let and operated as a tennis facility, with the tenancy understood to run until 2036; the grass courts host the qualifying tournament for The Championships, Wimbledon each June. The Grange itself has been retained by the Bank of England and forms the application site.

Roehampton lies beneath the approach paths to Heathrow Airport, and aircraft overflights are a prominent and regular feature of the noise climate throughout the day, particularly when the airport is operating on westerly arrivals. There are no significant roads in the immediate vicinity of the building; Bank Lane is a quiet residential lane, with Priory Lane and Roehampton Lane some distance to the east.

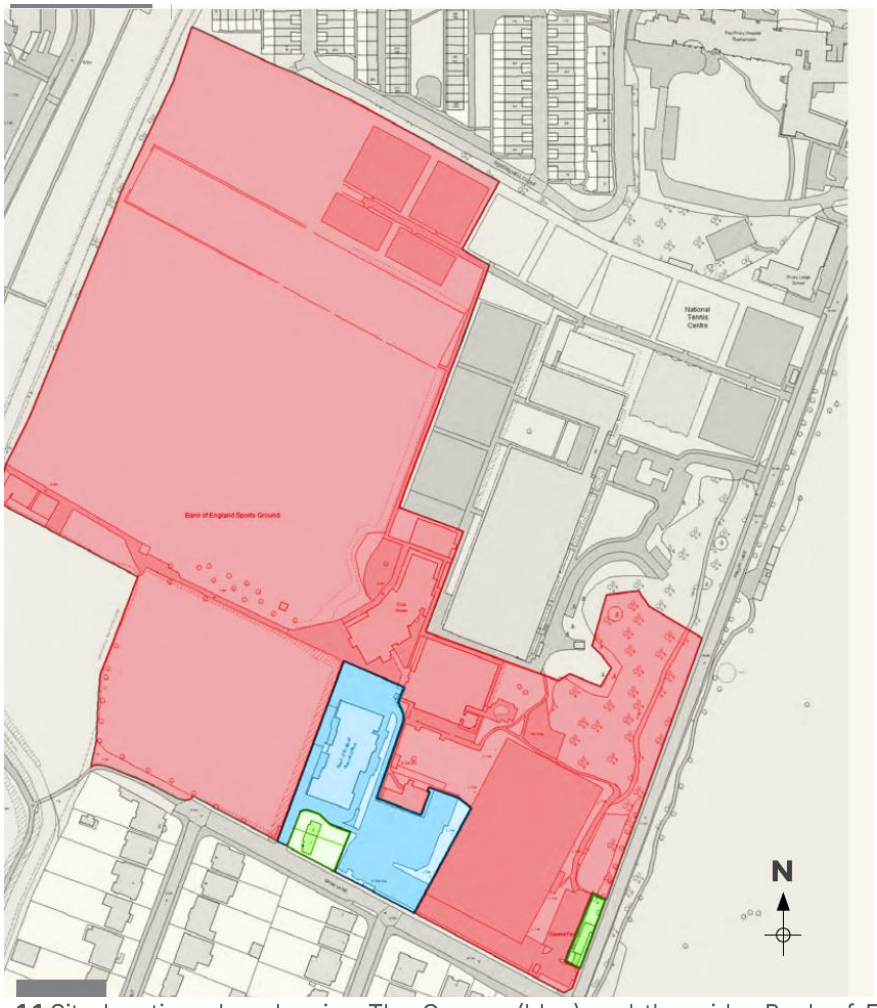


Figure 1.1 Site location plan showing The Grange (blue) and the wider Bank of England Sports Ground (red)



Figure 1.2 Aerial view of the sports ground looking north, with The Grange at the lower left, the pavilion and the National Tennis Centre beyond it, and the all-weather pitch to its south-east

1.3 Development Proposals

The proposals, as shown on the Michael Eales Architects drawings provided (393-200 to 207 Rev D02 dated 03 August 2026 and accommodation schedule 393-901 Rev 02 dated 27 August 2026), comprise the change of use of the whole of the existing building to provide 48No self-contained apartments over five floors (ground to fourth), comprising 5No studios, 17No one-bedroom, 19No two-bedroom and 7No three-bedroom apartments. Cycle and refuse stores are located at ground floor level and anticipated plant space at basement level. The existing external envelope of the building is retained; as the proposals are for a change of use under permitted development, the existing and proposed elevations are the same. The proposed floor plans are reproduced in Figure 1.3.

Habitable rooms are located on all elevations of the building, including the inner elevations of the two wings. No external fixed plant is proposed as part of the development and no commercial use is proposed to be retained within the building.

The design team has confirmed that the existing windows are to be retained, the alteration of the external windows falling outside the scope of Class MA. The existing windows are single glazed with secondary glazing installed internally to all habitable room windows, and no new or additional secondary glazing is proposed. Ventilation of the apartments is proposed to be by mechanical extract to the kitchens and bathrooms, with a general reliance on passive ventilation.



Figure 1.3 Proposed floor plans (reproduced from Michael Eales Architects drawing 393-200 Rev D02)

1.4 Assessment Scope

As the proposal is being progressed under permitted development (Class MA), the scope of the prior approval is limited to the matters specified in paragraph MA.2(2) of Schedule 2, Part 3 of the Town and Country Planning (General Permitted Development) (England) Order 2015, as amended. The relevant matter for this assessment is:

- impacts of noise from commercial premises on the intended occupiers of the development.

Accordingly, this assessment considers noise from the commercial premises surrounding the site, namely the sports ground and its pavilion, playing pitches and tennis courts, and the National Tennis Centre, as it affects the future occupiers of the proposed apartments, with reference to the internal ambient noise level guidance of BS 8233:2014 and the World Health Organization, and the principles of BS 4142:2014+A1:2019.

Noise from other sources, such as aircraft and road traffic, is not a relevant matter under the prior approval scope and should not be considered by the Local Planning Authority in determining the application. However, the surrounding sports uses are not conventional commercial premises, and the noise they generate is intermittent and variable in nature and cannot readily be separated from the other sources of noise affecting the site within a long-term measurement. For robustness, therefore, the assessment has been undertaken on the basis that the whole of the noise measured at the site throughout the survey period, from all sources, is treated as if it were noise from commercial premises, and the building envelope has been assessed against that total

noise climate, so that the internal ambient noise levels recommended by BS 8233 are achieved against it. This represents a considerable overstatement of the actual commercial noise affecting the site and provides a substantial margin of safety in respect of the relevant matter.

2.0 Assessment Guidance and Requirements

2.1 Permitted Development Requirements

Class MA of Part 3 of Schedule 2 to the General Permitted Development Order permits the change of use of a building and any land within its curtilage from a use falling within Class E (commercial, business and service) to a use falling within Class C3 (dwellinghouses), subject to the prior approval of the local planning authority as to a number of specified matters, including 'impacts of noise from commercial premises on the intended occupiers of the development'. The development is being progressed under these provisions, which require the London Borough of Wandsworth to consider that matter as part of the prior approval process.

In considering the impact of noise, the Local Planning Authority will have regard to the National Planning Policy Framework (NPPF) and the Noise Policy Statement for England (NPSE), which together require that new development should be appropriate for its location and that adverse impacts from noise on health and quality of life should be mitigated and reduced to a minimum, with significant adverse impacts avoided. The NPPF also establishes the 'agent of change' principle, under which new development which would be affected by an existing business or community facility should provide suitable mitigation so that the existing use does not have unreasonable restrictions placed upon it as a result of the development. The assessment has been prepared with these principles in mind, and the mitigation recommended is provided within the development itself.

2.2 BS 8233:2014

BS 8233:2014 'Guidance on sound insulation and noise reduction for buildings' (herein BS 8233) provides guidance on suitable internal ambient noise levels within residential buildings, along with recommended external amenity noise levels. The guidance draws on the World Health Organization (WHO) 'Guidelines for Community Noise' (1999) and 'Night Noise Guidelines for Europe' (2009). The internal ambient noise levels recommended for dwellings are set out in Table 2.1.

Table 2.1 Internal ambient noise levels for dwellings (BS 8233:2014)

Activity	Location	07:00 to 23:00	23:00 to 07:00
Resting	Living room	35 dB $L_{Aeq,16h}$	–
Dining	Dining room / area	40 dB $L_{Aeq,16h}$	–
Sleeping (daytime resting)	Bedroom	35 dB $L_{Aeq,16h}$	30 dB $L_{Aeq,8h}$

BS 8233 notes that where internal L_{Aeq} targets are achieved, occasional higher noise events may still cause disturbance and refers to WHO guidance on individual event levels. A commonly applied criterion, drawn from the WHO 'Guidelines for Community Noise', is that the internal L_{AFmax} in bedrooms should not regularly exceed 45 dB at night, this being applied as a level which should not be exceeded more than 10 to 15 times per night. The proposed studio apartments are single rooms used for both living and sleeping, and the bedroom criteria have been applied to them.

For external amenity areas such as gardens, balconies and terraces, BS 8233 states that it is desirable that the external noise level does not exceed 50 dB $L_{Aeq,T}$, with an upper guideline value of 55 dB $L_{Aeq,T}$ which would be acceptable in noisier environments. It recognises that these values are not achievable in all circumstances, that in higher-noise areas a compromise between elevated noise levels and other factors may be warranted, and that development should not be prohibited on noise grounds alone.

2.3 BS 4142:2014+A1:2019

BS 4142:2014+A1:2019 'Methods for rating and assessing industrial and commercial sound' (herein BS 4142) provides a method for assessing the likelihood of adverse impact from industrial and commercial sound sources at nearby noise-sensitive receptors. The assessment compares the rating level of the source in question against the prevailing background sound level at the receptor location. The rating level is derived from the measured or predicted specific sound level, with corrections applied where the sound has distinctive characteristics such as tonality, impulsivity, or intermittency.

The difference between the rating level and the background sound level provides an initial estimate of impact. A difference of around +10 dB or more is likely to be an indication of a significant adverse impact, depending on context. A difference of around +5 dB is likely to be an indication of an adverse impact, depending on context. Where the rating level does not exceed the background sound level, this is an indication of the specific sound source having a low impact, depending on context. These initial estimates are then considered in context, taking account of factors such as the absolute noise levels, the character of the existing acoustic environment, the sensitivity of the receptor, and the nature and frequency of the sound.

BS 4142 is intended primarily for the assessment of fixed industrial and commercial installations and is of limited direct application to the intermittent noise of outdoor sports activity. It has been used in this assessment to characterise the prevailing background sound levels at the site, rather than as the determining method; the determining assessment is of the internal ambient noise levels within the proposed apartments against the guidance of BS 8233 and the WHO.

3.0 Environmental Noise Survey

3.1 Procedure

Continuous automated noise measurements were undertaken at two measurement positions on the existing building (MP1 and MP2) over a period of seven days, from 11:15 on 12 August 2026 to 13:15 on 19 August 2026.

The sound level meters were set to log the L_{Aeq} and L_{AFmax} at one-minute intervals, and to store the L_{Aeq}, L_{AFmax}, L_{AFmin} and statistical (L_{An}) levels, together with the octave band L_{Zeq}, L_{ZFmax} and L_{ZFmin} spectra from 16 Hz to 16 kHz, over consecutive 15-minute periods. The measurements captured all noise affecting the building from all sources, including activity on the sports ground, the National Tennis Centre and the all-weather pitch, aircraft, road traffic and any existing building services plant.

Weather conditions during the survey were dry with light winds and therefore suitable for the measurement of environmental noise. The measurement procedure complied with ISO 1996-2:2017 'Acoustics – Description, measurement and assessment of environmental noise – Part 2: Determination of sound pressure levels'.

3.2 Noise Climate

The noise climate at the site is dominated by aircraft. The time histories in Appendix A show the characteristic diurnal pattern of Heathrow arrivals: a quiet period in the early hours with a background sound level in the low 30s dB L_{A90} at both positions, a marked increase from around 04:30 to 05:00 when the first arrivals of the day pass over, sustained overflights at intervals of one to two minutes through the daytime at 65 to 80 dB L_{AFmax}, and a change in the level and frequency of overflights in the mid-afternoon or evening on several days consistent with the alternation of the arrival runways. The individual overflights govern the daytime and night-time L_{Aeq} and the L_{AFmax} levels at both positions; the daytime background sound level of around 40 dB L_{A90} between overflights is set by distant road traffic and the general activity of the area.

Activity on the sports ground, on the tennis courts and at the National Tennis Centre during the survey was only just audible at the building during the daytime and early evening as intermittent voices, ball impacts and the occasional whistle or shout, at levels which are not separately distinguishable from the aircraft-dominated time histories. The pavilion is no longer in use for functions.

3.3 Measurement Positions

The measurement positions are shown in Figure 3.1. MP1 was located at the north-eastern corner of the building, at a second-floor window facing the National Tennis Centre and the pavilion to the north-east, and MP2 on the eastern elevation of the building at a second-floor window facing the car park and the all-weather pitch to the east and south-east. The two positions therefore face the commercial premises of principal interest; the western elevations of the building, facing the playing fields and Bank Lane, are exposed to the same overhead aircraft noise and are more distant from the pitch and the National Tennis Centre. Both microphones were mounted approximately 1 m from the façade, extended from the window, and the measured levels have been corrected by -3 dB to free-field conditions for the purposes of the assessment. The measured levels at the two positions are closely similar, reflecting the overhead nature of the dominant source.

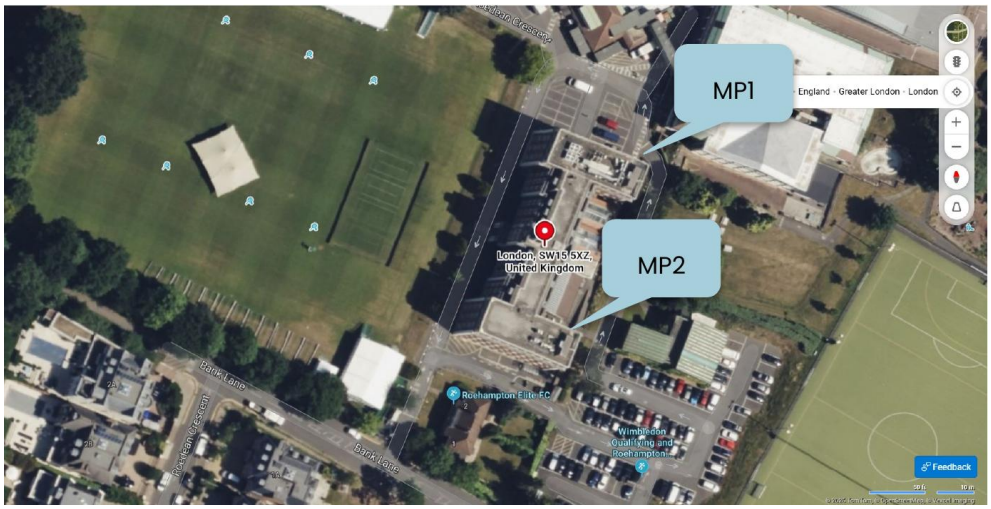


Figure 3.1 Environmental noise survey measurement positions

3.4 Equipment

The equipment used during the survey is summarised in Table 3.1. The equipment calibration was verified before and after use and no abnormalities were observed.

Table 3.1 Environmental noise survey equipment

Equipment	Serial No.	Calibration Date	Certificate Ref.
Class 1 Sound Level Meter, Svantek Type 971A	173136	19 Jan 2026	1516680-1
Preamplifier, Svantek SV18A	163637	19 Jan 2026	1516680-1
Free-field Microphone, ACO 7152	96767	19 Jan 2026	1516680-1
Class 1 Sound Level Meter, Svantek Type 971A	173141	19 Jan 2026	1516680-2
Preamplifier, Svantek SV18A	163626	19 Jan 2026	1516680-2
Free-field Microphone, ACO 7152	96756	19 Jan 2026	1516680-2
Class 1 Acoustic Calibrator, Svantek SV33B	182123	30 Jan 2026	C182123-1

3.5 Results

The results for each daytime and night-time period throughout the survey are presented for MP1 in Table 3.2 and for MP2 in Table 3.3, with the levels adopted for the assessment in the final row of each table. Time-history graphs of the survey and the distribution of the measured $L_{A90,15min}$ levels at each position are presented in Appendix A. All levels in this report are free-field sound levels unless stated otherwise, with a -3 dB correction applied to the measured façade data.

Table 3.2 Environmental noise survey results, MP1

Date	Daytime (07:00 – 23:00)		Night-time (23:00 – 07:00)		
	$L_{Aeq,16h}$ (dB)	$L_{A90,15min}$ (dB)	$L_{Aeq,8h}$ (dB)	$L_{A90,15min}$ (dB)	L_{AFmax} (dB)
Wed 12 August	45	38	37	30	57
Thu 13 August	52	39	46	31	66
Fri 14 August	52	39	38	32	53
Sat 15 August	43	35	46	29	63
Sun 16 August	52	36	49	28	69
Mon 17 August	54	38	49	32	70
Tue 18 August	59	39	50	28	70
Adopted	**59**	**36**	**50**	**28**	**70**

Table 3.3 Environmental noise survey results, MP2

Date	Daytime (07:00 – 23:00)		Night-time (23:00 – 07:00)		
	$L_{Aeq,16h}$ (dB)	$L_{A90,15min}$ (dB)	$L_{Aeq,8h}$ (dB)	$L_{A90,15min}$ (dB)	L_{AFmax} (dB)
Wed 12 August	43	39	39	34	54
Thu 13 August	54	45	49	44	68
Fri 14 August	53	45	45	45	54
Sat 15 August	46	45	49	43	66
Sun 16 August	53	44	50	30	71
Mon 17 August	55	39	50	31	71
Tue 18 August	57	39	51	30	71
Adopted	**57**	**39**	**51**	**30**	**71**

4.0 Commercial Noise Assessment

4.1 Commercial Premises Surrounding the Site

The commercial premises in the vicinity of the site, all of which are sports and leisure uses, are as follows:

- **the sports pavilion (clubhouse)**, now disused, approximately 30 m to the north of the building, together with the adjoining sports hall with swimming pool, gym and studios, approximately 15 m to the north-east of the building;
- **the all-weather (2G) sports pitch**, approximately 65 m to the south-east of the building beyond the car park, used for football and training;
- **the grass playing fields, cricket squares and tennis courts** of the sports ground, which adjoin the building to the west and extend to the north and north-west, with grass courts laid out on the playing field from approximately 20 m from the western elevation and hard courts approximately 250 m to the north, now operated as a tennis facility and hosting the Wimbledon qualifying tournament for two weeks each June;
- **the National Tennis Centre** of the Lawn Tennis Association to the north-east, comprising indoor and outdoor courts, the nearest part of which (its indoor tennis hall) is approximately 60 m from the building beyond the pavilion and sports hall, and the **Roehampton Club** golf course beyond it; and
- **the three stand-alone squash courts** approximately 15 m to the east of the building, adjacent to the car park, which are disused.

The term 'commercial premises' is not defined in the General Permitted Development Order. The sports ground, pavilion and National Tennis Centre are non-residential premises operated on a business basis, and they have therefore been treated as commercial premises for the purposes of the prior approval so that the Local Planning Authority may be satisfied that the relevant matter has been fully addressed. There are no other commercial premises within the vicinity of the building; the surrounding streets are wholly residential. No commercial use is retained within The Grange itself under the proposals.

Noise from outdoor sports activity is intermittent and variable in character: voices, shouting and whistles, ball impacts on hard courts and on the all-weather pitch, and occasional public address or crowd noise during tournament weeks, generally confined to the daytime and early evening.

4.2 Assessment Approach

The noise from the surrounding sports uses could not be isolated from the other sources of noise affecting the site within the survey data, being intermittent and considerably lower in level than the aircraft overflights which dominate the measured L_{Aeq} and L_{AFmax} at both positions. Rather than attempt to apportion the measured levels between sources, the assessment has been undertaken on the robust basis described in Section 1.4: the total measured noise climate at the site, from all sources and on the noisiest day and night of the survey, has been adopted as the design noise level for the building envelope, and the sound insulation and ventilation requirements of the envelope have been derived so that the internal ambient noise level criteria of BS 8233 and the WHO are achieved against that total noise climate.

This approach necessarily overstates the noise from commercial premises by a wide margin, since the adopted levels are governed by aircraft.

4.3 Criteria

Based on the guidance set out within BS 8233 and the WHO guidelines, the internal ambient noise levels of Table 2.1, together with an internal level of 45 dB L_{AFmax} in bedrooms at night not to be exceeded more than 10 to 15 times per night (Section 2.2), have been adopted for the proposed apartments. As the studio apartments are single rooms used for both living and sleeping, the bedroom criteria have been applied to them.

The design noise levels adopted for all elevations of the building are the higher of the two measured positions: 59 dB $L_{Aeq,16h}$ daytime, 51 dB $L_{Aeq,8h}$ night-time and 71 dB L_{AFmax} for regular night-time events (free-field), equivalent to 62 dB, 54 dB and 74 dB respectively at the façade. As the dominant source is overhead and the two positions differ by no more than 2 dB, a single specification has been applied to all elevations of the building, including the inner elevations of the wings. The octave band spectra measured at the governing position for each parameter (MP1 for the daytime $L_{Aeq,16h}$ and MP2 for the night-time $L_{Aeq,8h}$ and L_{AFmax}) have been used in the façade calculations of Section 4.6.

4.4 Internal Noise Levels with Windows Open

A partially open window provides a level difference of the order of 13 to 15 dB between the external façade level and the internal ambient level. With windows open, the internal levels within the apartments would be of the order of 47 dB $L_{Aeq,16h}$ during the daytime, 39 dB $L_{Aeq,8h}$ during the night-time and 59 dB L_{AFmax} for regular night-time events. These exceed the criteria of Section 4.3, and it is not therefore possible to rely on open windows for the ventilation of habitable rooms whilst achieving the internal ambient noise criteria against the total noise climate. The criteria are achieved with windows closed and the existing windows and secondary glazing retained, as set out below.

4.5 Non-Glazed Elements

The retained external walls of the existing building are of solid masonry construction of substantial thickness. For the purposes of the façade calculations the walls have conservatively been taken as 215 mm solid brick, plastered, with the sound insulation performance shown in Table 4.1; the existing walls are expected to be thicker and to exceed this performance, and in all cases they comfortably exceed the performance of the glazed elements. All external constructions should be free of gaps and unsealed penetrations, and any new openings formed in the envelope, including those for ventilators and extract ducts, should be sealed to the same standard.

Table 4.1 Assumed sound reduction performance for non-glazed elements

Element	Sound Reduction Performance (R dB) at Octave Band Centre Frequency (Hz)						$R_w + C_{tr}$ (dB)
	125	250	500	1k	2k	4k	
Retained solid masonry external wall	41	44	48	55	55	55	49

4.6 Glazed Elements

The existing windows are to be retained, and are single glazed with secondary glazing installed internally to all habitable room windows. The secondary glazing is to be retained, and no new or additional secondary glazing is proposed. The existing windows comprise 4 mm single glazing with secondary glazing of 6 mm glass at a cavity of approximately 100 mm.

The internal noise levels with windows closed have been calculated using a façade sound insulation calculation to the method of BS EN 12354-3, as set out in Annex G of BS 8233. The sound reduction of the glazing and the external wall is combined as a composite sound reduction index for the façade, the sound transmitted through the background ventilators is added as a separate path using their element-normalised level difference ($D_{n,e}$), and the internal level is derived from the free-field incident level, the façade area and the absorption of the room at each octave band from 63 Hz to 8 kHz. The incident levels are the octave band spectra measured at the governing position for each parameter (Section 4.3), and a reverberation time of 0.5 s has been assumed for furnished rooms.

Two rooms have been assessed: a typical bedroom of 3.5 m by 4.0 m with 3.0 m² of glazing in an 8.75 m² façade, and, as a worst case, a corner bedroom of 3.5 m by 3.5 m with 4.8 m² of glazing across two elevations in a total façade area of 12.25 m². The glazed areas are based on the existing window openings shown on the measured survey elevations, of approximately 0.8 m and 1.4 m width by 2.0 m height, and the room sizes on the proposed plans; a room height of 2.5 m has been assumed, which is conservative

given the floor-to-floor heights of the existing building. The larger studio apartments and living rooms have greater room volumes relative to their glazed areas and are less onerous.

To represent the case where background ventilators are installed in the façade, the calculations include 2No ventilators per room at the performance set out in Section 4.7. For the avoidance of doubt, the sound insulation data used are shown in Table 4.2.

Table 4.2 Sound insulation data used in the façade calculations

Element	Sound Reduction Performance (dB) at Octave Band Centre Frequency (Hz)						$R_w + C_{tr}$ (dB)
	125	250	500	1k	2k	4k	
Existing single glazing (4 mm)	20	22	25	31	34	29	27
Existing single glazing with secondary glazing (4 mm / 100 mm cavity / 6 mm)	26	34	44	56	53	52	37
External wall (as 215 mm solid brick, plastered)	41	44	48	55	55	55	49
Background ventilator at the minimum specification, per unit ($D_{n,e}$; single figure $D_{n,e,w} + C_{tr}$)	41	38	34	33	33	33	34

The predicted internal noise levels are set out in Table 4.3 for the existing windows with their secondary glazing and, as a sensitivity test, for standard single glazing alone, i.e. with the secondary glazing removed.

Table 4.3 Predicted internal noise levels with windows closed (dB)

Scenario	Daytime $L_{Aeq,16h}$	Night-time $L_{Aeq,8h}$	Night-time L_{AFmax}
Criterion (bedrooms and studio apartments)	**35**	**30**	**45**
Existing windows with secondary glazing: typical bedroom	31	23	43
Existing windows with secondary glazing: corner bedroom	32	24	44
Sensitivity test, single glazing only: typical bedroom	*33*	*26*	*46*
Sensitivity test, single glazing only: corner bedroom	*34*	*27*	*47*

With the existing windows and secondary glazing retained, the internal criteria are achieved in all rooms against the total noise climate. The night-time L_{AFmax}, the most

onerous parameter, is 43 dB in the typical bedroom and 44 dB in the corner bedroom, within the criterion of 45 dB, and the daytime and night-time L_{Aeq} are within the criteria by 3 dB or more.

In this configuration the glazing contributes only a small part of the internal level, which is controlled by the background ventilators where these are installed; without ventilators the calculated internal L_{AFmax} would be of the order of 31 to 33 dB. No additional secondary glazing or replacement glazing is therefore required.

With the secondary glazing omitted, the night-time L_{AFmax} would exceed the criterion by around 1 dB in the typical bedroom (46 dB) and by around 2 dB in the corner bedroom (47 dB), although single glazing alone would achieve the daytime criterion of 35 dB $L_{Aeq,16h}$ in the living rooms, with a calculated level of around 31 to 33 dB. The secondary glazing is therefore relied upon to achieve the night-time criteria in the bedrooms and studio apartments, and it should be retained to all bedroom and studio windows; its retention to the living rooms is also recommended. It is reiterated that the night-time criteria are governed by aircraft on the approach to Heathrow which is not a commercial use, and that the sports uses do not operate at night. **The retention of the secondary glazing should not therefore be a determining factor under the permitted development rights.**

4.7 Ventilation

4.7.1 Normal Ventilation

The design team has confirmed that the apartments are to be provided with mechanical extract ventilation to the kitchens and bathrooms, with a general reliance on passive ventilation. Based on the noise levels measured on site, appropriate ventilation systems (ref. Approved Document F, ADF) are outlined in Table 4.4 in order that the internal noise criteria are not compromised.

Table 4.4 Ventilation strategy

Ventilation System (ADF)	Whole Dwelling Ventilation	Extract Ventilation
System 1	Background ventilators providing a minimum performance as described below	Intermittent extract fans
System 3	Continuous mechanical extract (low rate) and background ventilators for supply providing a minimum performance as described below	Continuous mechanical extract (high rate) with background ventilators providing inlet air
System 4	Continuous mechanical supply and extract (low rate)	Continuous mechanical supply and extract (high rate)

Where background ventilators are to be installed in the façade of habitable rooms, they should achieve a minimum performance of **$D_{n,e,w}$ + C_{tr} 34 dB** where **2No** ventilators are provided per room, as included in the calculations of Section 4.6, or **$D_{n,e,w}$ + C_{tr} 31 dB** where a single ventilator is provided per room. Increasing the number of ventilators will reduce the composite performance of the façade; if more than 2No ventilators are required, the required performance of each should be increased by 10·log(N/2) dB, where N is the number of ventilators proposed per room.

In the case of mechanical ventilation, systems should be designed to meet the internal noise levels as defined in CIBSE Guide A (2015), as shown in Table 4.5. In all cases, purge ventilation may be provided by openable windows; the internal noise level requirement would not be applicable during purge conditions as this would only occur occasionally. Opening the secondary glazing as well as the window would be required for purge ventilation.

Table 4.5 Mechanical building services noise level requirements

Room Type	L_{Aeq}, dB	NR
Bedrooms and studio apartments	30	25
Living rooms	35	30
Kitchens	45-50	40-45

4.7.2 Overheating (Approved Document O)

Approved Document O 'Overheating' (herein ADO) applies to new residential buildings and does not apply to dwellings formed by a material change of use, **and an assessment against ADO is not therefore required for the development**. For information, the noise guidance of ADO concerns the night-time period only (23:00 to 07:00), during which windows are likely to be closed if the internal level would exceed 40 dB $L_{Aeq,8h}$, or 55 dB L_{AFmax} more than 10 times per night. The sports uses surrounding the site do not operate during the night-time period and the pavilion is no longer in use as a function venue, and noise from commercial premises would not therefore prevent windows from being opened at night for the removal of excess heat, in line with ADO. Against the total noise climate, the internal level with windows open would be approximately 39 dB $L_{Aeq,8h}$, within the ADO value, but regular aircraft events would give approximately 59 dB L_{AFmax}, above it; this is governed by aircraft on the approach to Heathrow rather than by the commercial premises, and does not form part of the prior approval.

4.8 Existing Building Services Plant

It is understood that some of the existing roof-mounted plant serving the offices remains in operation and that the existing boiler installation may be retained. These installations form part of the building itself and would serve the residential use; they are not commercial premises and are not a relevant matter under the prior approval. Any plant retained to serve the apartments, and any new plant installed, should be selected, located and attenuated so that its rating level in accordance with BS 4142 does not exceed the prevailing background sound level at the nearest habitable room window of the development and of the existing dwellings on Bank Lane, taken as the adopted background sound levels in Tables 3.2 and 3.3, and so that the internal noise levels of Table 4.5 are achieved within the apartments. Plant serving the offices which is no longer required should be decommissioned and removed.

4.9 External Amenity

No private gardens or balconies are proposed. The external areas around the building are subject to daytime levels of 59 dB $L_{Aeq,16h}$ on the noisiest day of the survey and 54 dB $L_{Aeq,16h}$ averaged across the survey, governed by aircraft. These levels are at or marginally above the BS 8233 upper guideline value of 55 dB $L_{Aeq,16h}$, which in a location beneath the Heathrow approach paths is considered an acceptable compromise in accordance with the guidance of BS 8233, and the extensive parkland and open space of the surrounding area provides ample alternative amenity. This is provided for information and is not a relevant matter under the prior approval.

5.0 Conclusion

Murray Acoustics Ltd has undertaken a noise impact assessment of the proposed change of use of The Grange, Bank Lane, Roehampton, from offices to 48No residential apartments, in support of a prior approval application to the London Borough of Wandsworth under Class MA of the Town and Country Planning (General Permitted Development) (England) Order 2015 (as amended).

The relevant matter under the prior approval scope is the impact of noise from commercial premises on the intended occupiers of the development. The commercial premises in the vicinity of the site are the sports ground and its pavilion, playing pitches and tennis courts, and the National Tennis Centre to the north-east. A seven-day environmental noise survey was undertaken at two positions on the building from 12 to 19 August 2026, capturing all noise affecting the site. The noise climate is dominated by aircraft on the approach to Heathrow; noise from the surrounding sports uses was audible during the daytime as intermittent voices and ball impacts, but was not separately distinguishable from the aircraft-dominated noise climate.

As the sports uses are not conventional commercial premises and their noise cannot be separated from the other sources within the survey, the assessment has been undertaken on the robust basis that the total measured noise climate (59 dB $L_{Aeq,16h}$, 51 dB $L_{Aeq,8h}$ and 71 dB L_{AFmax}, free-field) is treated as noise from commercial premises, and the building envelope has been assessed so that the internal ambient noise levels recommended by BS 8233 and the WHO are achieved against that total noise climate. This considerably overstates the commercial noise affecting the site and provides a substantial margin of safety.

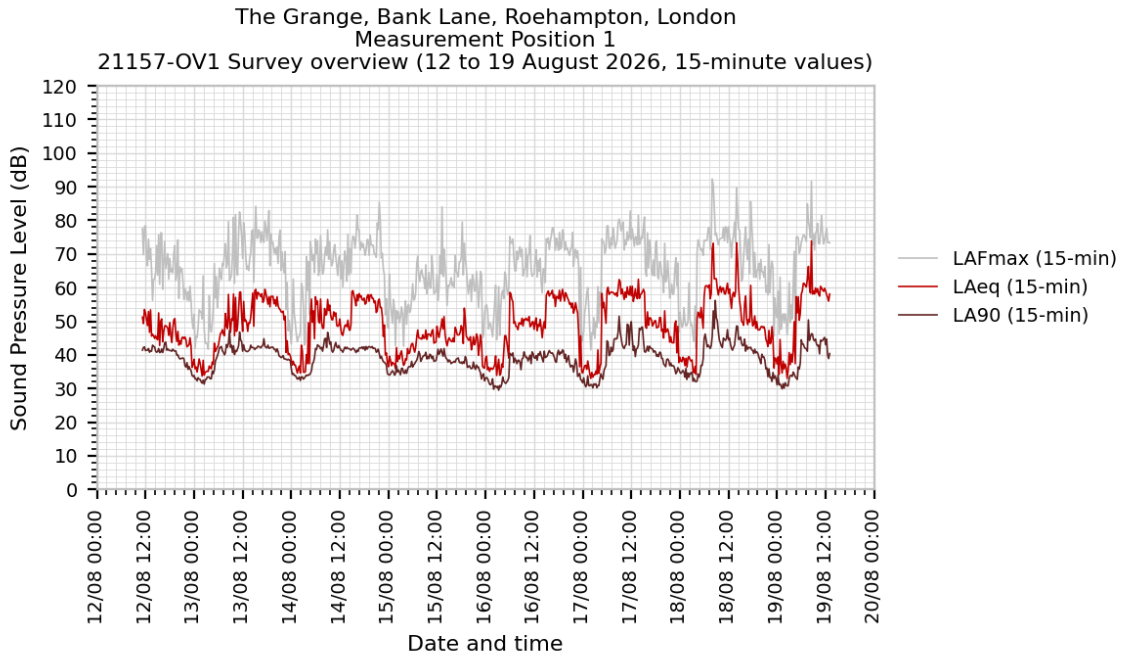
The existing windows, which are single glazed with secondary glazing installed internally to all habitable rooms, are to be retained. A façade sound insulation calculation shows that, with the existing windows and secondary glazing retained, and with any background ventilators installed in the façade achieving $D_{n,e,w}$ + C_{tr} 34 dB (2No per room) or 31 dB (1No per room), the internal criteria are achieved with windows closed in all habitable rooms against the total noise climate. No additional or replacement glazing is required, and the existing secondary glazing should be retained and maintained in good condition. Without the secondary glazing, the night-time L_{AFmax} criterion would be exceeded by around 1 to 2 dB; this is governed by aircraft on the approach to Heathrow, which is not a commercial use, and not by the commercial premises, which do not operate at night. The retention of the secondary glazing should not therefore be a determining factor under the permitted development rights.

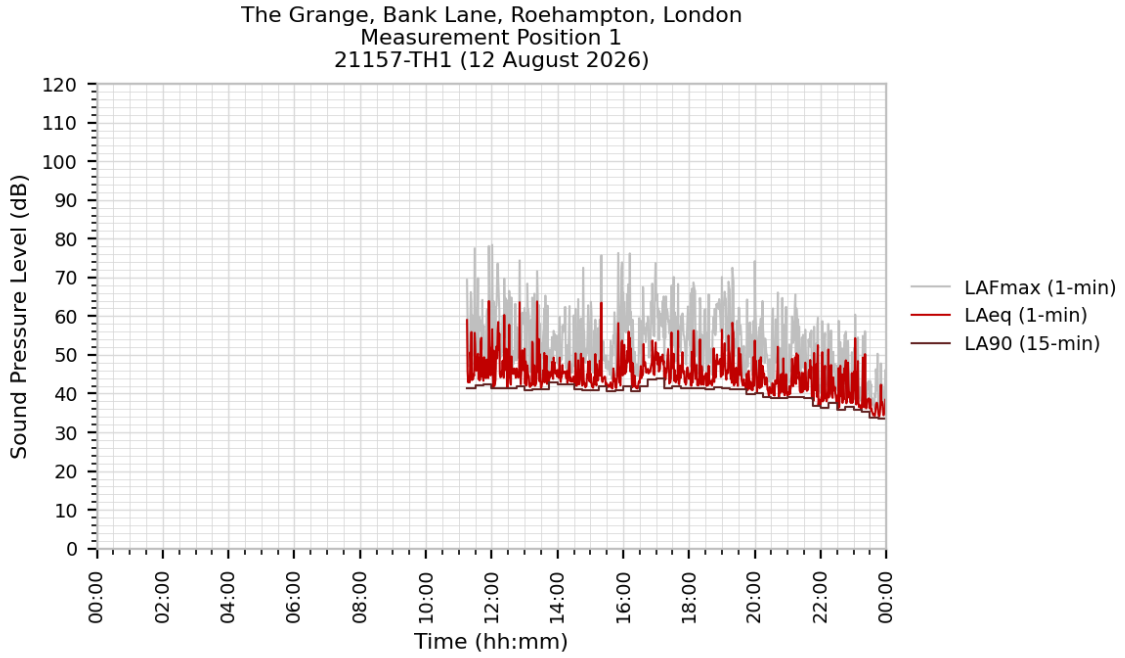
Approved Document O does not apply to the change of use. The commercial premises surrounding the site do not operate at night, and noise from commercial premises would not prevent windows from being opened at night for the removal of excess heat. Any

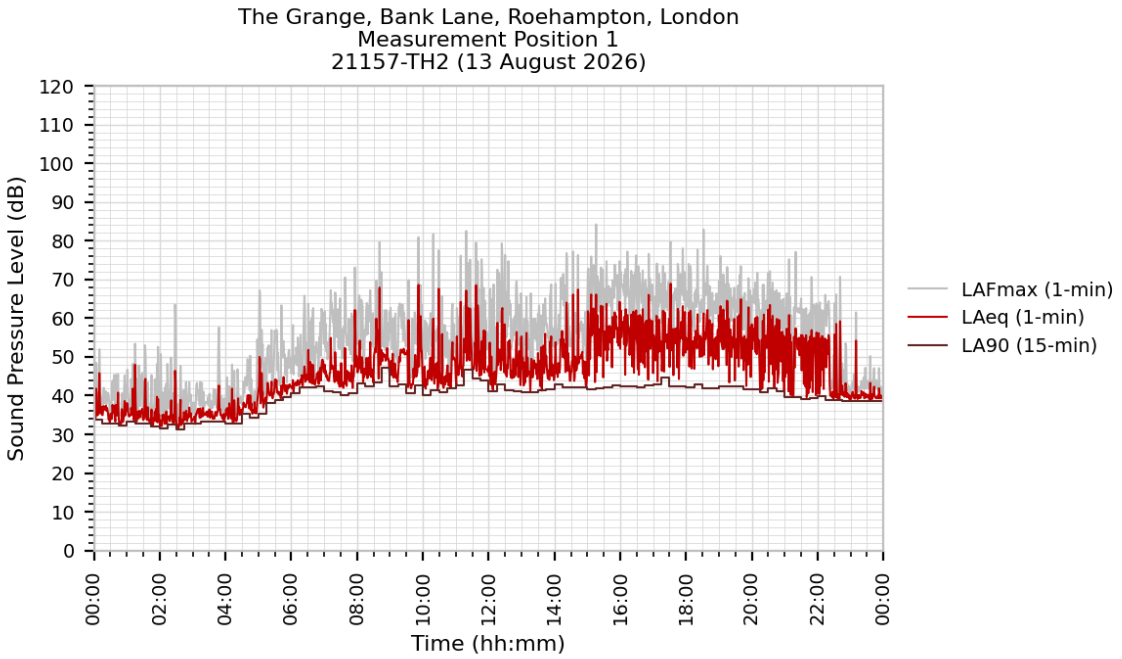
existing plant retained to serve the apartments, and any new plant, should be designed to the limits set out in Section 4.8.

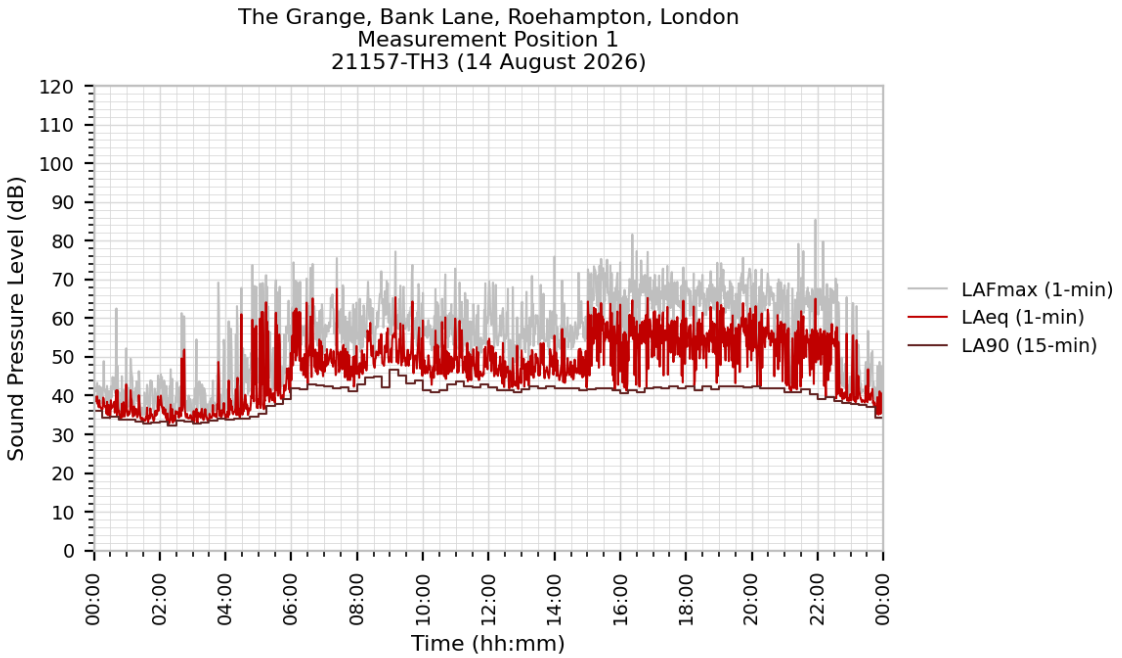
Noise from the surrounding sports uses is intermittent and confined to the daytime and early evening, and with the retained building envelope the internal levels arising from the commercial premises would be well within the adopted criteria. Noise from commercial premises is therefore not considered to give rise to an adverse impact on the future occupiers of the development. The proposed change of use is acceptable in noise terms with regard to the relevant matter under the prior approval scope, and prior approval is recommended on noise grounds.

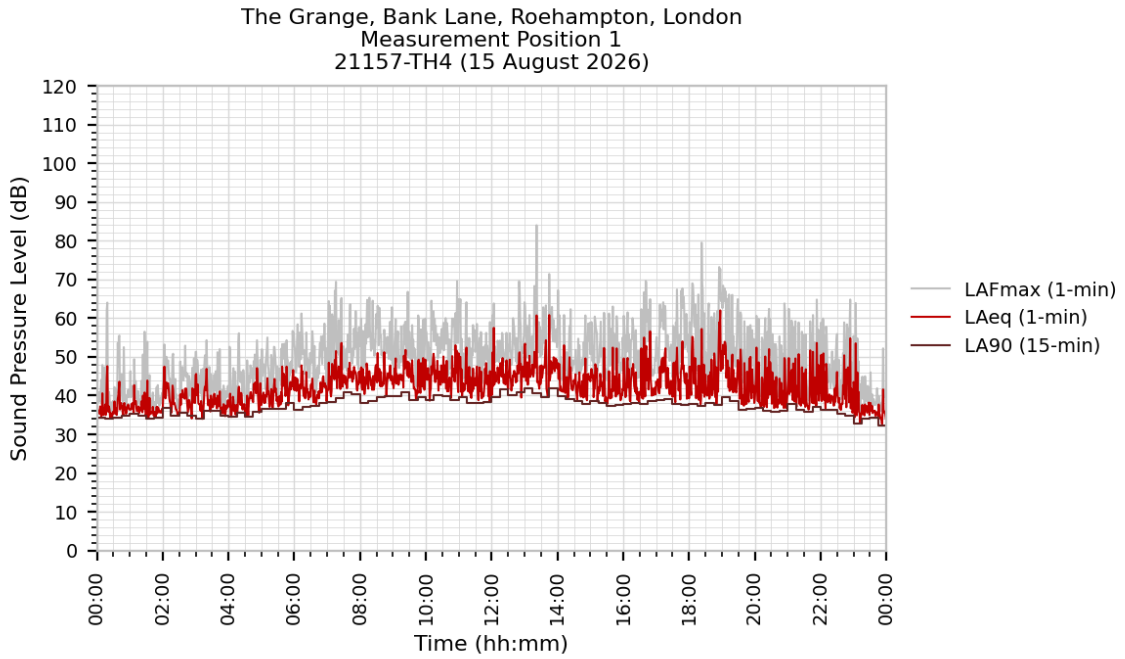
Appendix A – Environmental Noise Survey Results

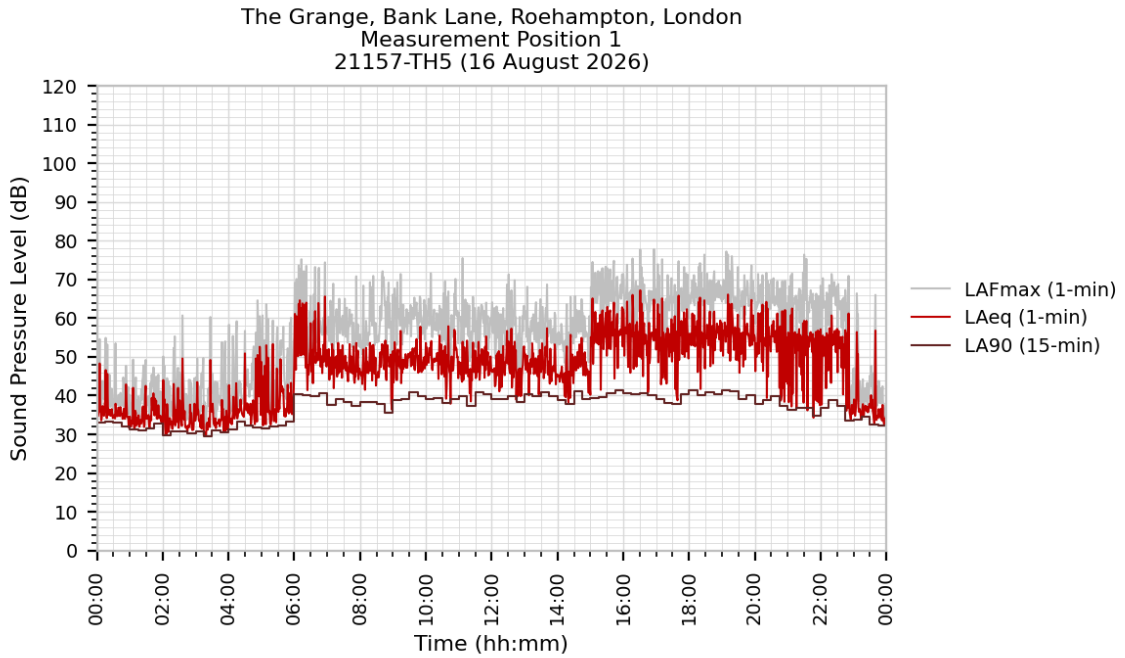


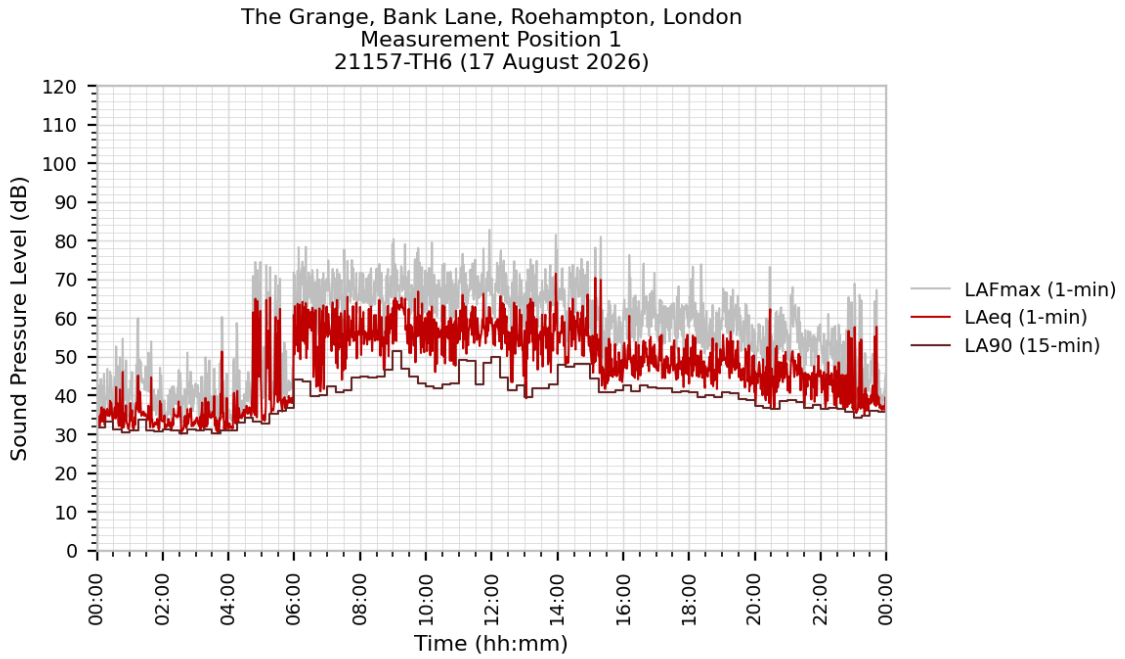


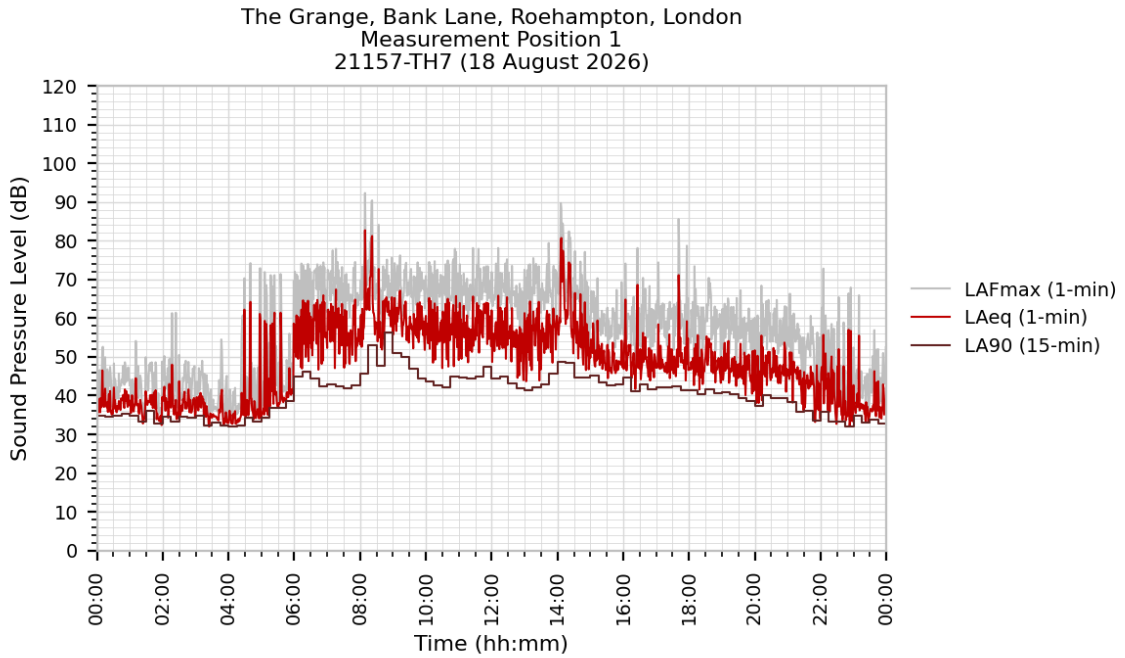

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-TH2 (13 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

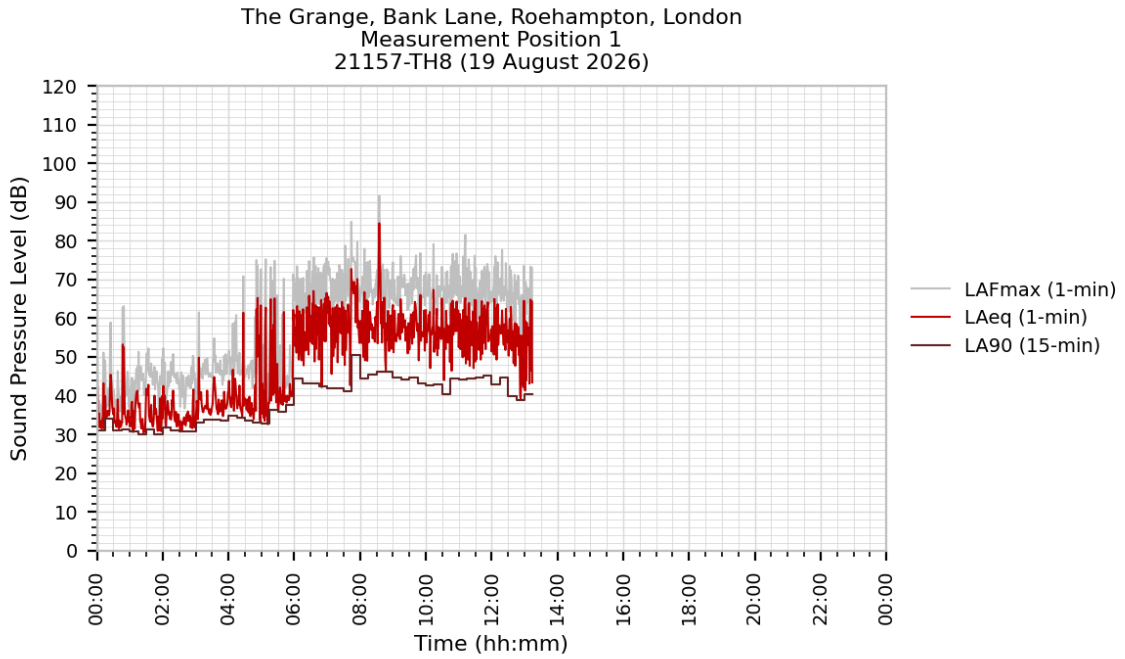

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-TH3 (14 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

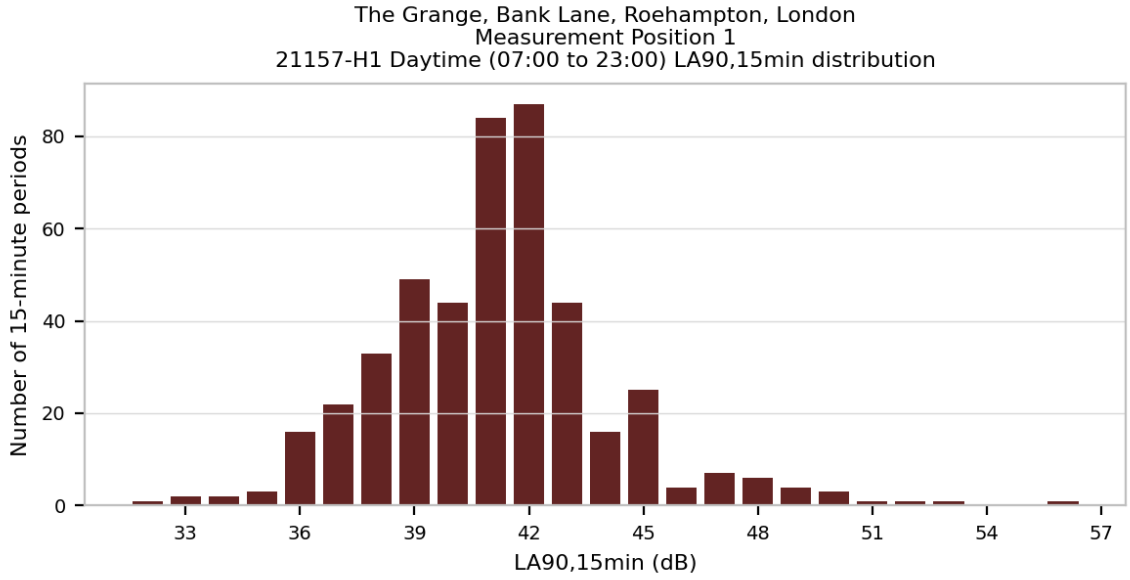

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-TH4 (15 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

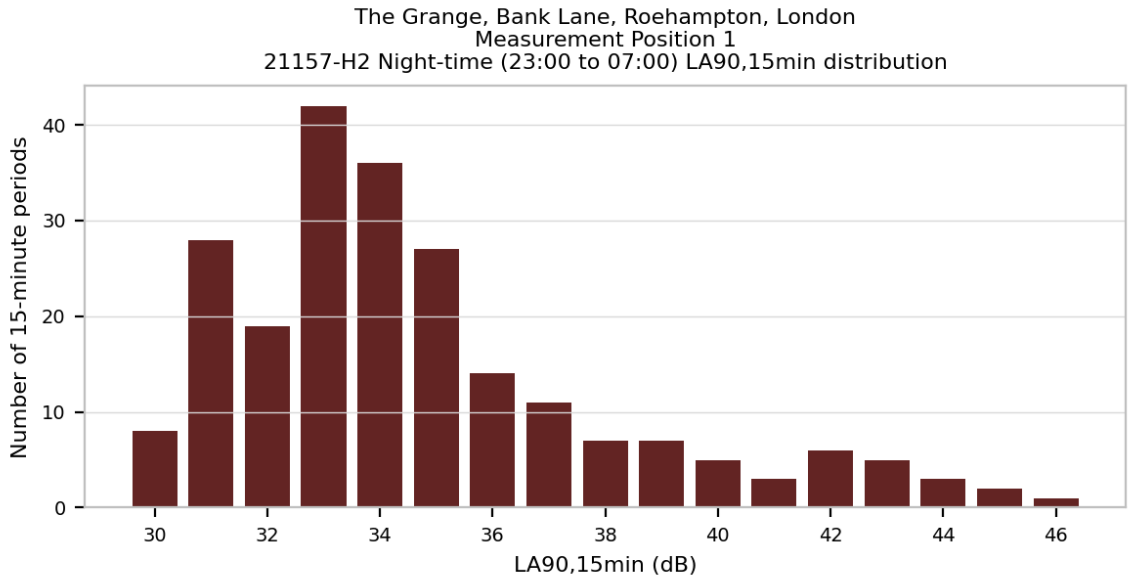

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-TH5 (16 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

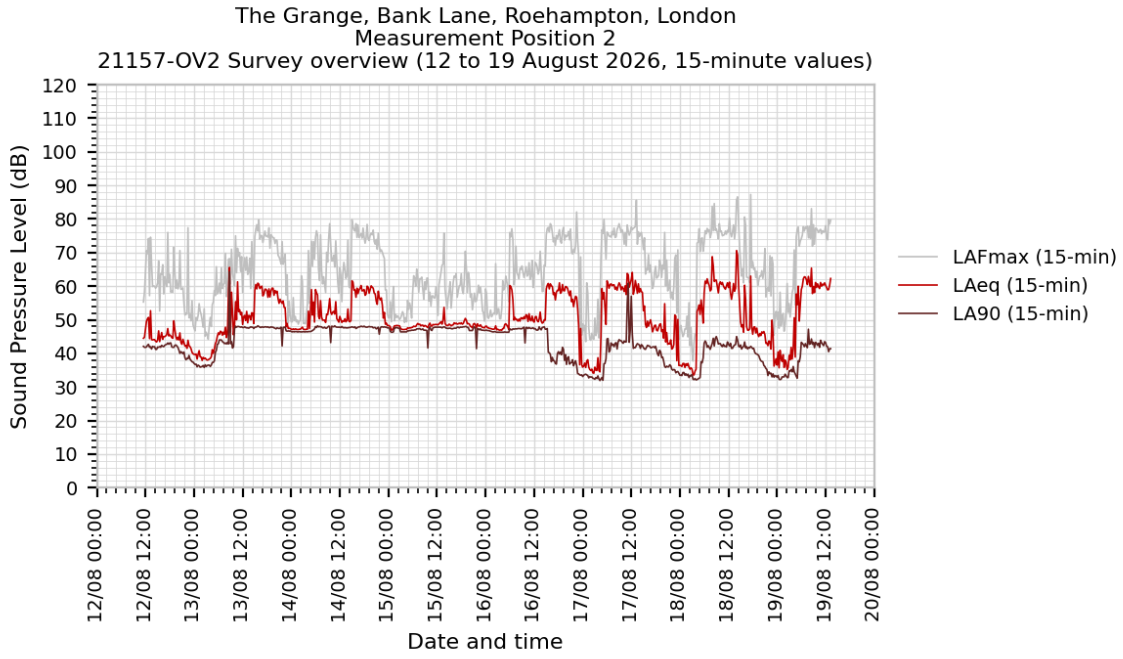

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-TH6 (17 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

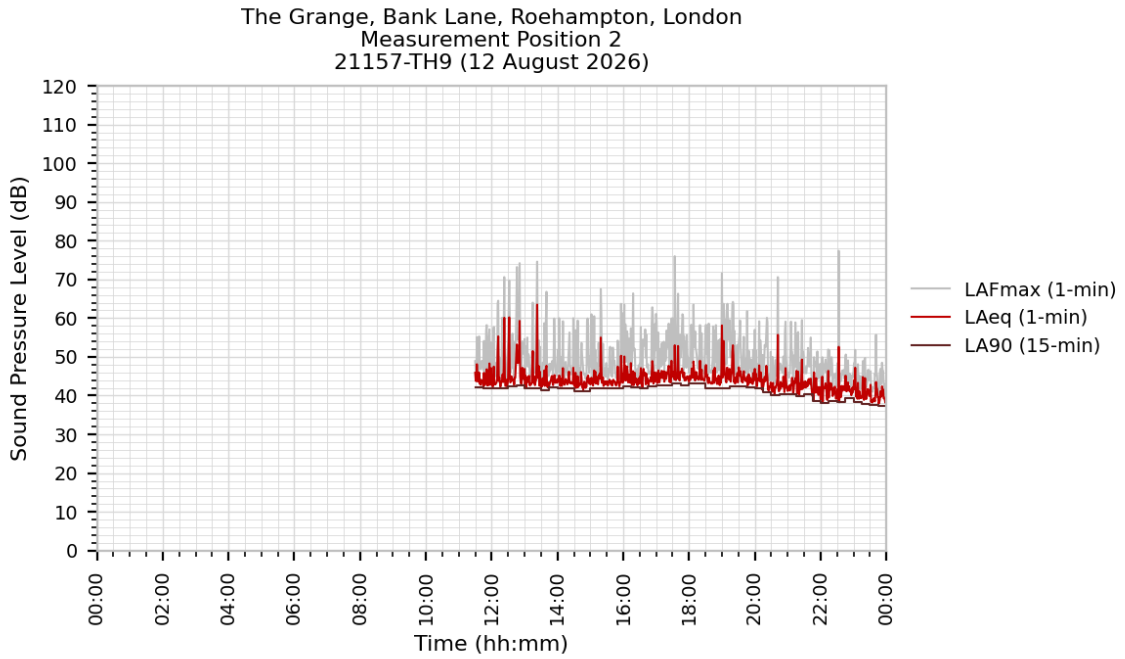

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-TH7 (18 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

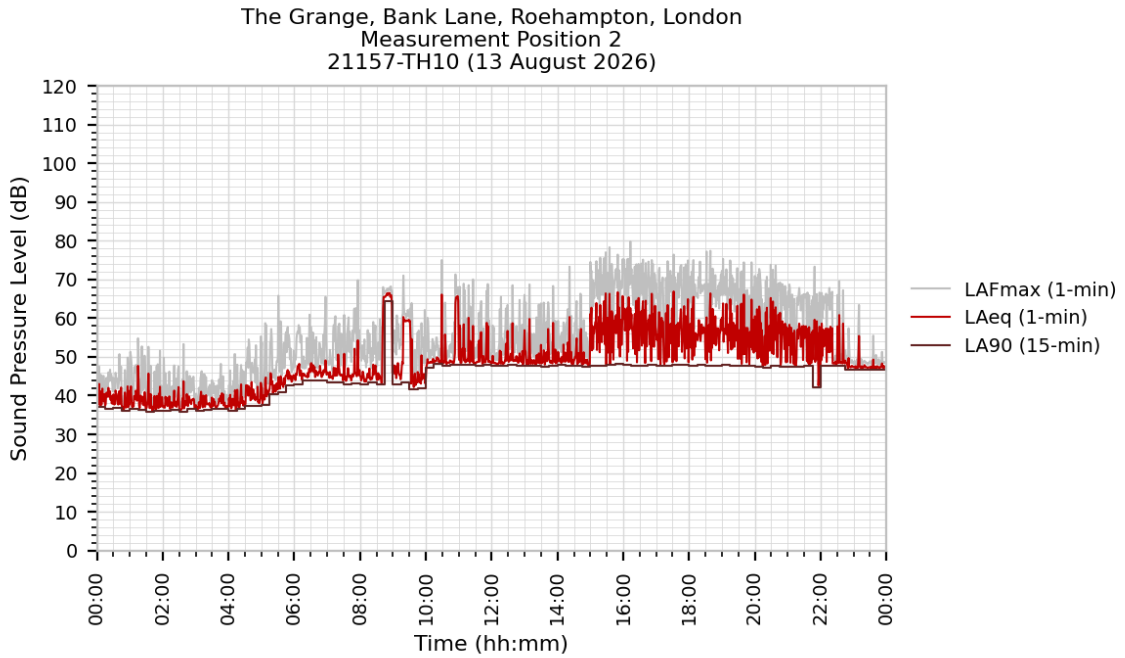

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-TH8 (19 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

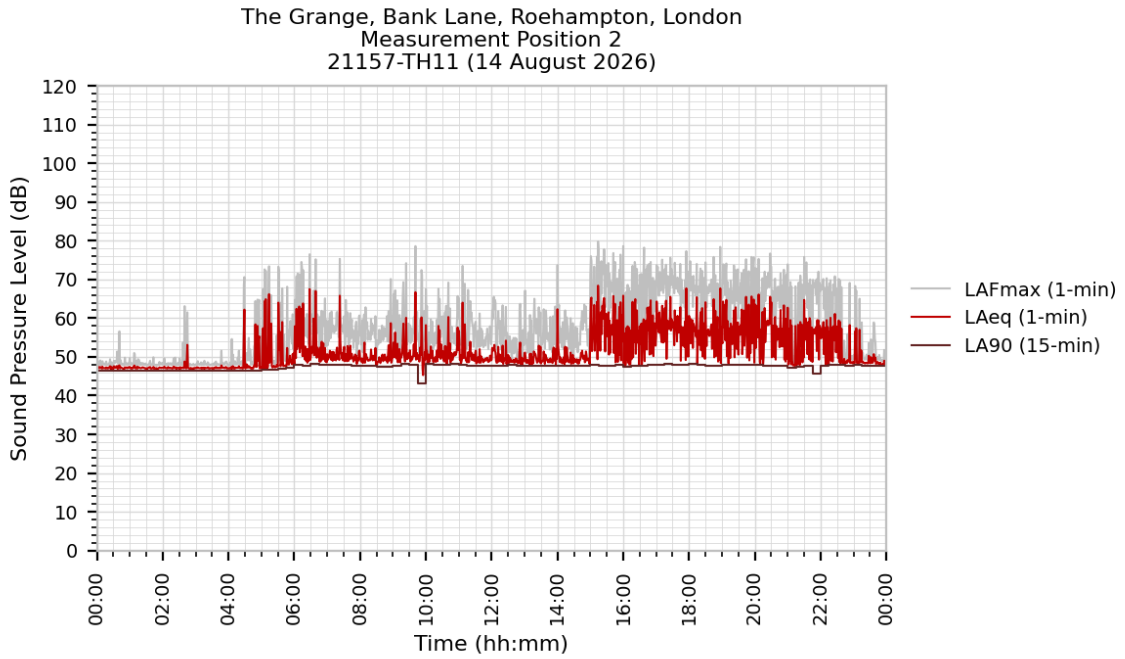

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-H1 Daytime (07:00 to 23:00) LA90,15min distribution
Number of 15-minute periods
80
60
40
20
0
33
36
39
42
45
48
51
54
57
LA90,15min (dB)

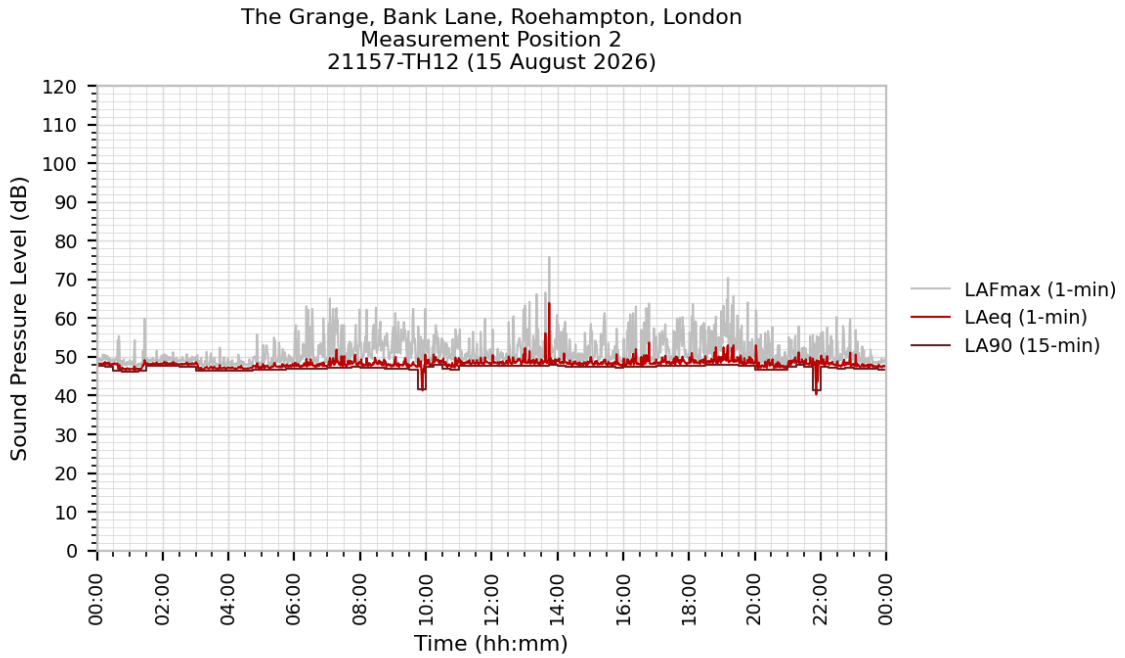

The Grange, Bank Lane, Roehampton, London
Measurement Position 1
21157-H2 Night-time (23:00 to 07:00) LA90,15min distribution
Number of 15-minute periods
0
10
20
30
40
30
32
34
36
38
40
42
44
46
LA90,15min (dB)

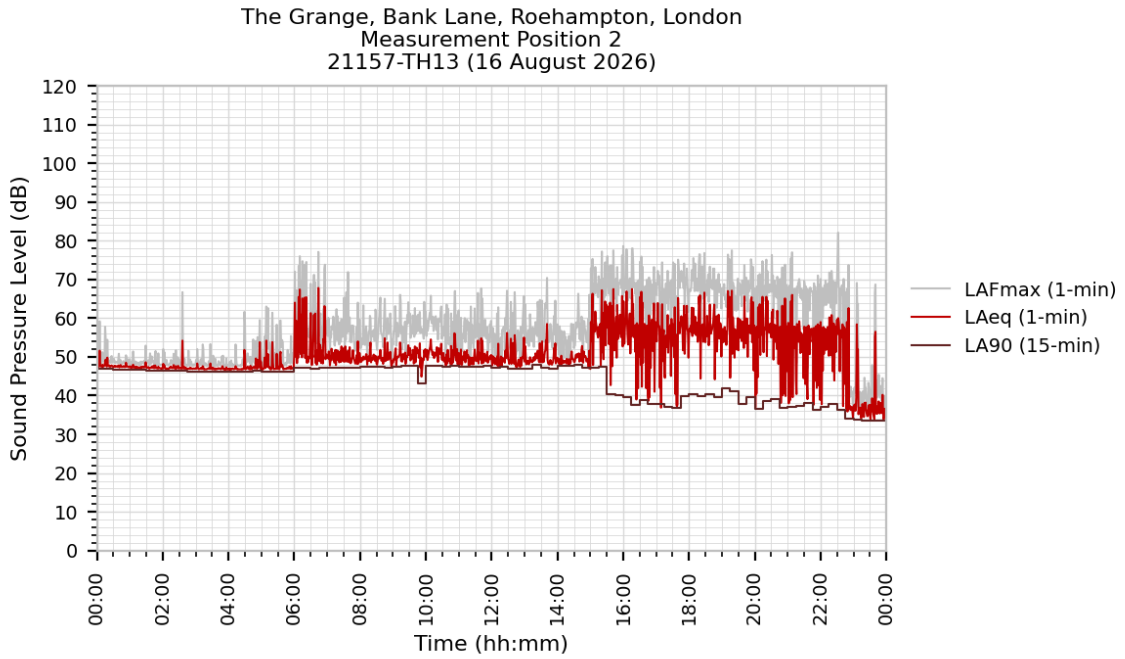

The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-OV2 Survey overview (12 to 19 August 2026, 15-minute values)
Sound Pressure Level (dB)
0
10
20
30
40
50
60
70
80
90
100
110
120
12/08 00:00
12/08 12:00
13/08 00:00
13/08 12:00
14/08 00:00
14/08 12:00
15/08 00:00
15/08 12:00
16/08 00:00
16/08 12:00
17/08 00:00
17/08 12:00
18/08 00:00
18/08 12:00
19/08 00:00
19/08 12:00
20/08 00:00
Date and time
LAFmax (15-min)
LAeq (15-min)
LA90 (15-min)

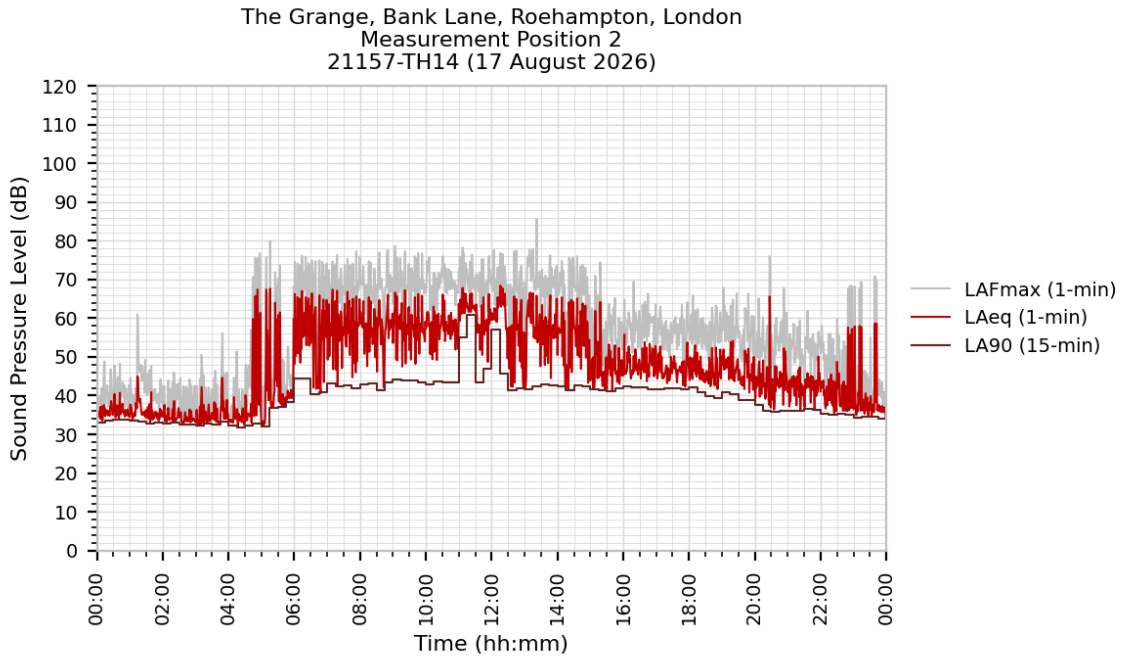

The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH9 (12 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

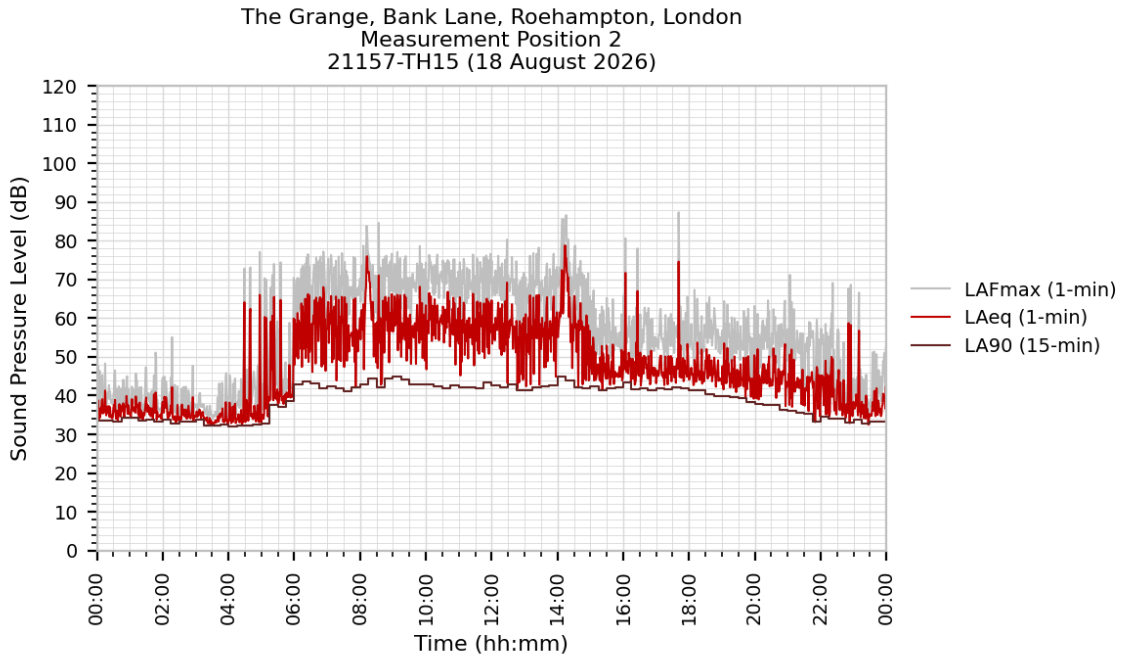

The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH10 (13 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

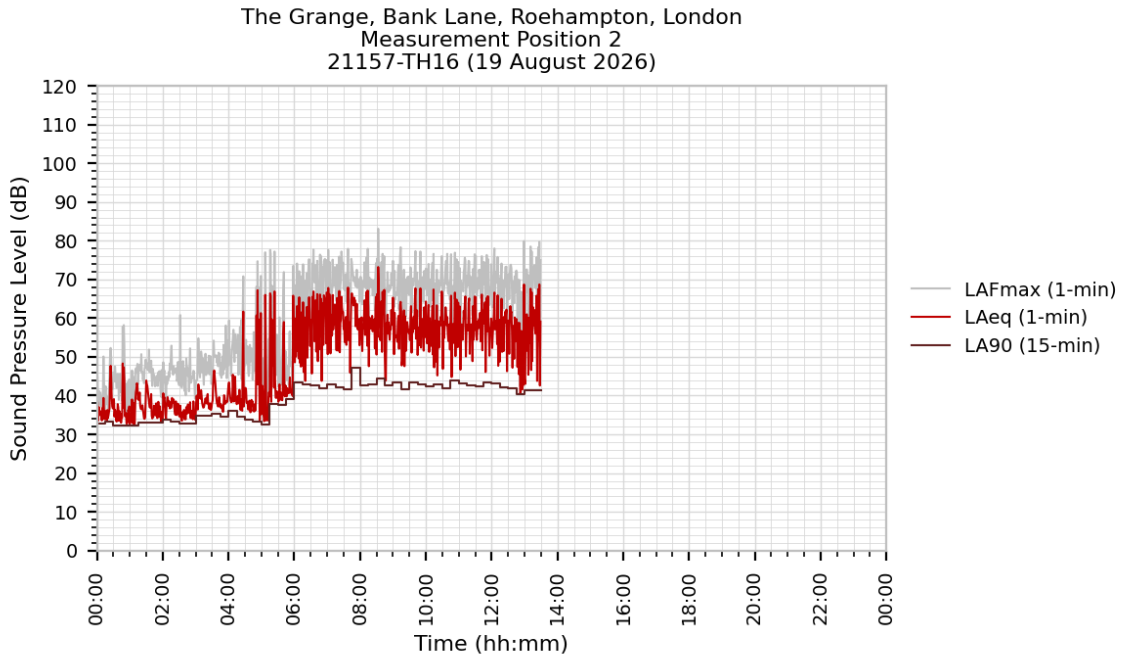

The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH11 (14 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

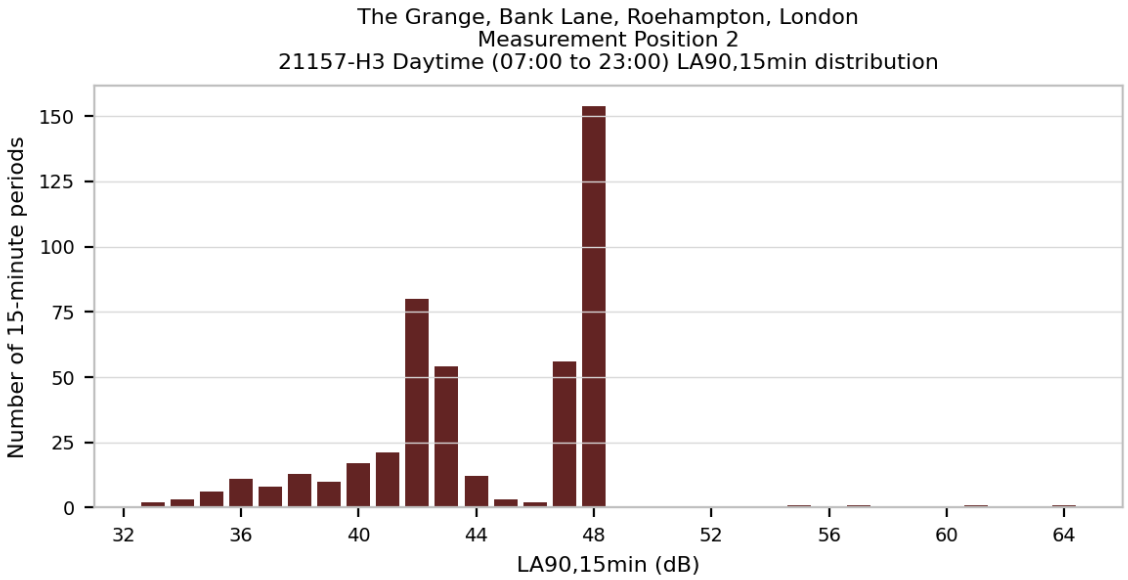

The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH12 (15 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)

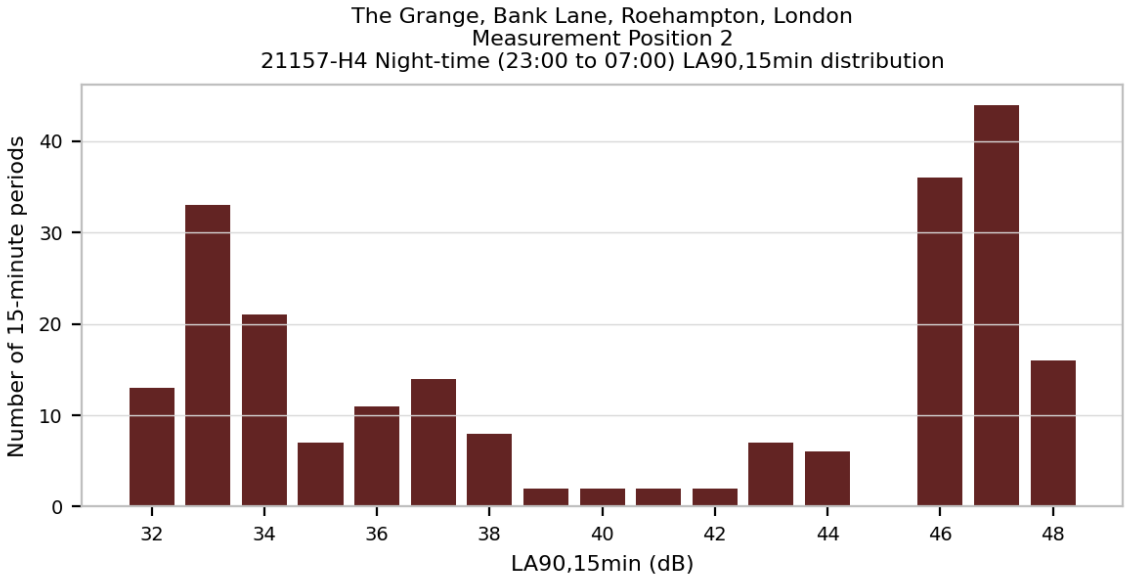

The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH13 (16 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)


The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH14 (17 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)


The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH15 (18 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)


The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-TH16 (19 August 2026)
Sound Pressure Level (dB)
120
110
100
90
80
70
60
50
40
30
20
10
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
Time (hh:mm)
LAFmax (1-min)
LAeq (1-min)
LA90 (15-min)


The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-H3 Daytime (07:00 to 23:00) LA90,15min distribution
Number of 15-minute periods
0
25
50
75
100
125
150
32
36
40
44
48
52
56
60
64
LA90,15min (dB)


The Grange, Bank Lane, Roehampton, London
Measurement Position 2
21157-H4 Night-time (23:00 to 07:00) LA90,15min distribution
Number of 15-minute periods
0
10
20
30
40
32
34
36
38
40
42
44
46
48
LA90,15min (dB)